

Oracle HR Schema JOIN Practice Task

1. HR Schema Relationship Analysis

In the Oracle HR schema, employee data is connected through foreign keys.

Main relationships

EMPLOYEES.department_id → DEPARTMENTS.department_id

EMPLOYEES.job_id → JOBS.job_id

DEPARTMENTS.location_id → LOCATIONS.location_id

LOCATIONS.country_id → COUNTRIES.country_id

COUNTRIES.region_id → REGIONS.region_id

Business meaning

Each employee has a job.

Each employee may belong to a department.

Each department is located in a location.

Each location belongs to a country.

Each country belongs to a region.

Query 1: Employee Job Report

Business requirement

Show each employee with their job title.

Correct SQL

```
SELECT
```

```
    emp.employee_id,
```

```
    emp.first_name,
```

```
    emp.last_name,
```

```
    job.job_title
```

FROM employees emp

JOIN jobs job

ON emp.job_id = job.job_id

ORDER BY emp.employee_id;

Worksheet

Query Builder

```
SELECT
    emp.employee_id,
    emp.first_name,
    emp.last_name,
    job.job_title
FROM employees emp
JOIN jobs job
    ON emp.job_id = job.job_id
ORDER BY emp.employee_id;
```

Script Output x

Query Result x

SQL | Fetched 50 rows in 0.006 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	JOB_TITLE
1	100	Steven	King	President
2	101	Neena	Kochhar	Administration Vice President
3	102	Lex	De Haan	Administration Vice President
4	103	Alexander	Hunold	Programmer
5	104	Bruce	Ernst	Programmer
6	105	David	Austin	Programmer
7	106	Valli	Pataballa	Programmer
8	107	Diana	Lorentz	Programmer
9	108	Nancy	Greenberg	Finance Manager
10	109	Daniel	Faviet	Accountant
11	110	John	Chen	Accountant
12	111	Ismael	Sciarra	Accountant
13	112	Jose Manuel	Urman	Accountant
14	113	Luis	Popp	Accountant
15	114	Den	Raphaely	Purchasing Manager
16	115	Alexander	Khoo	Purchasing Clerk
17	116	Shelli	Baida	Purchasing Clerk
18	117	Sigal	Tobias	Purchasing Clerk
19	118	Guy	Himuro	Purchasing Clerk
20	119	Karen	Colmenares	Purchasing Clerk
21	120	Matthew	Weiss	Stock Manager
22	121	Adam	Finn	Stock Manager

Explanation

The correct relationship is:

employees.job_id = jobs.job_id

Do **not** join `employee_id` with `job_id` because they are different business keys.

Query 2: Employee Department Report

Business requirement

Show employees with their department names, including employees without a department.

Correct SQL

```
SELECT
    emp.employee_id,
    emp.first_name,
    emp.last_name,
    dept.department_name
FROM employees emp
LEFT JOIN departments dept
    ON emp.department_id = dept.department_id
ORDER BY emp.employee_id;
```

Worksheet

Query Builder

```

SELECT
    emp.employee_id,
    emp.first_name,
    emp.last_name,
    dept.department_name
FROM employees emp
LEFT JOIN departments dept
    ON emp.department_id = dept.department_id
ORDER BY emp.employee_id;

```

Script Output x

Query Result x

SQL | Fetched 50 rows in 0.006 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME
1	100	Steven	King	Executive
2	101	Neena	Kochhar	Executive
3	102	Lex	De Haan	Executive
4	103	Alexander	Hunold	IT
5	104	Bruce	Ernst	IT
6	105	David	Austin	IT
7	106	Valli	Pataballa	IT
8	107	Diana	Lorentz	IT
9	108	Nancy	Greenberg	Finance
10	109	Daniel	Faviet	Finance
11	110	John	Chen	Finance
12	111	Ismael	Sciarra	Finance
13	112	Jose Manuel	Urman	Finance
14	113	Luis	Popp	Finance
15	114	Den	Raphaely	Purchasing
16	115	Alexander	Khoo	Purchasing
17	116	Shelli	Baida	Purchasing
18	117	Sigal	Tobias	Purchasing
19	118	Guy	Himuro	Purchasing
20	119	Karen	Colmenares	Purchasing
21	120	Matthew	Weiss	Shipping
22	121	Adam	Fraier	Shipping

Explanation

LEFT JOIN is used because some employees may not be assigned to a department.

Query 3: Sales Department Employees

Business requirement

Show employees who work in the Sales department.

Correct SQL

```
SELECT
    emp.employee_id,
    emp.first_name,
    emp.last_name,
    dept.department_name
FROM employees emp
JOIN departments dept
    ON emp.department_id = dept.department_id
WHERE dept.department_name = 'Sales'
ORDER BY emp.employee_id;
```

Worksheet

Query Builder

```
SELECT
    emp.employee_id,
    emp.first_name,
    emp.last_name,
    dept.department_name
FROM employees emp
JOIN departments dept
    ON emp.department_id = dept.department_id
WHERE dept.department_name = 'Sales'
ORDER BY emp.employee_id;
```

Script Output x

Query Result x

SQL | All Rows Fetched: 34 in 0.006 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME
1	145	John	Russell	Sales
2	146	Karen	Partners	Sales
3	147	Alberto	Errazuriz	Sales
4	148	Gerald	Cambrault	Sales
5	149	Eleni	Zlotkey	Sales
6	150	Peter	Tucker	Sales
7	151	David	Bernstein	Sales
8	152	Peter	Hall	Sales
9	153	Christopher	Olsen	Sales
10	154	Nanette	Cambrault	Sales
11	155	Oliver	Tuvault	Sales
12	156	Janette	King	Sales
13	157	Patrick	Sully	Sales
14	158	Allan	McEwen	Sales
15	159	Lindsey	Smith	Sales
16	160	Louise	Doran	Sales
17	161	Sarath	Sewall	Sales
18	162	Clara	Vishney	Sales
19	163	Danielle	Greene	Sales
20	164	Mattea	Marvins	Sales
21	165	David	Lee	Sales
22	166	Sundar	Ade	Sales

Explanation

Use **JOIN** here because the business requirement asks only for employees who belong to the Sales department.

Using this:

LEFT JOIN departments dept

WHERE dept.department_name = 'Sales'

turns the **LEFT JOIN** into an **INNER JOIN**, because the **WHERE** condition removes rows where department is **NULL**.

Query 4: Employee Manager Report

Business requirement

Show each employee with their manager name.

Correct SQL

SELECT

emp.employee_id,

emp.first_name AS employee_first_name,

emp.last_name AS employee_last_name,

mgr.first_name AS manager_first_name,

mgr.last_name AS manager_last_name

FROM employees emp

LEFT JOIN employees mgr

ON emp.manager_id = mgr.employee_id

ORDER BY emp.employee_id;

Worksheet

Query Builder

```
SELECT
emp.employee_id,
emp.first_name AS employee_first_name,
emp.last_name AS employee_last_name,
mgr.first_name AS manager_first_name,
mgr.last_name AS manager_last_name
FROM employees emp
LEFT JOIN employees mgr
ON emp.manager_id = mgr.employee_id
ORDER BY emp.employee_id;
```

Script Output x

Query Result x

SQL | Fetched 50 rows in 0.01 seconds

	EMPLOYEE_ID	EMPLOYEE_FIRST_NAME	EMPLOYEE_LAST_NAME	MANAGER_FIRST_NAME	MANAGER_LAST_NAME
1	100	Steven	King	(null)	(null)
2	101	Neena	Kochhar	Steven	King
3	102	Lex	De Haan	Steven	King
4	103	Alexander	Hunold	Lex	De Haan
5	104	Bruce	Ernst	Alexander	Hunold
6	105	David	Austin	Alexander	Hunold
7	106	Valli	Pataballa	Alexander	Hunold
8	107	Diana	Lorentz	Alexander	Hunold
9	108	Nancy	Greenberg	Neena	Kochhar
10	109	Daniel	Faviet	Nancy	Greenberg
11	110	John	Chen	Nancy	Greenberg
12	111	Ismael	Sciarra	Nancy	Greenberg
13	112	Jose Manuel	Urman	Nancy	Greenberg
14	113	Luis	Popp	Nancy	Greenberg
15	114	Den	Raphaely	Steven	King
16	115	Alexander	Khoo	Den	Raphaely
17	116	Shelli	Baida	Den	Raphaely
18	117	Sigal	Tobias	Den	Raphaely
19	118	Guy	Himuro	Den	Raphaely
20	119	Karen	Colmenares	Den	Raphaely
21	120	Matthew	Weiss	Steven	King
22	121	Adam	Frain	Steven	King

Explanation

This is a self join.

Correct relationship:

`emp.manager_id = mgr.employee_id`

Incorrect logic:

`emp.employee_id = mgr.manager_id`

That means you are finding employees who manage other employees, not showing each employee's manager.

LEFT JOIN is better because the top manager may not have a manager.

Query 5: Full Employee Location Report

Business requirement

Show employee details with department, job, city, country, and region.

Correct SQL

```
SELECT
    emp.employee_id,
    emp.first_name,
    emp.last_name,
    job.job_title,
    dept.department_name,
    loc.city,
    ctry.country_name,
    reg.region_name
FROM employees emp
JOIN jobs job
    ON emp.job_id = job.job_id
LEFT JOIN departments dept
    ON emp.department_id = dept.department_id
LEFT JOIN locations loc
    ON dept.location_id = loc.location_id
LEFT JOIN countries ctry
```

ON loc.country_id = ctry.country_id

LEFT JOIN regions reg

ON ctry.region_id = reg.region_id

ORDER BY emp.employee_id;

Worksheet

Query Builder

```
SELECT
emp.employee_id,
emp.first_name,
emp.last_name,
job.job_title,
dept.department_name,
loc.city,
ctry.country_name,
reg.region_name
FROM employees emp
JOIN jobs job
ON emp.job_id = job.job_id
LEFT JOIN departments dept
ON emp.department_id = dept.department_id
LEFT JOIN locations loc
ON dept.location_id = loc.location_id
LEFT JOIN countries ctry
ON loc.country_id = ctry.country_id
LEFT JOIN regions reg
ON ctry.region_id = reg.region_id
ORDER BY emp.employee_id;
```

Script Output x

Query Result x

SQL | Fetched 50 rows in 0.013 seconds

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	JOB_TITLE	DEPARTMENT_NAME	CITY	COUNTRY_NAME	REGION_NAME
1	Steven	King	President	Executive	Seattle	United States of America	Americas
2	Neena	Kochhar	Administration Vice President	Executive	Seattle	United States of America	Americas
3	Lex	De Haan	Administration Vice President	Executive	Seattle	United States of America	Americas
4	Alexander	Hunold	Programmer	IT	Southlake	United States of America	Americas
5	Bruce	Ernst	Programmer	IT	Southlake	United States of America	Americas
6	David	Austin	Programmer	IT	Southlake	United States of America	Americas
7	Valli	Pataballa	Programmer	IT	Southlake	United States of America	Americas
8	Diana	Lorentz	Programmer	IT	Southlake	United States of America	Americas
9	Nancy	Greenberg	Finance Manager	Finance	Seattle	United States of America	Americas
10	Daniel	Faviet	Accountant	Finance	Seattle	United States of America	Americas
11	John	Chen	Accountant	Finance	Seattle	United States of America	Americas
12	Ismael	Sciarra	Accountant	Finance	Seattle	United States of America	Americas
13	Jose Manuel	Urman	Accountant	Finance	Seattle	United States of America	Americas
14	Luis	Popp	Accountant	Finance	Seattle	United States of America	Americas

Explanation

This query follows the full HR data flow:

Employee → Job

Employee → Department

Department → Location

Location → Country

Country → Region

LEFT JOIN is used for department/location/country/region because some employees may not have department data.

Query 6: Department Location Report

Business requirement

Show departments with their city, country, and region.

Correct SQL

```
SELECT
    dept.department_id,
    dept.department_name,
    loc.city,
    ctry.country_name,
    reg.region_name
FROM departments dept
JOIN locations loc
    ON dept.location_id = loc.location_id
JOIN countries ctry
    ON loc.country_id = ctry.country_id
JOIN regions reg
    ON ctry.region_id = reg.region_id
ORDER BY dept.department_id;
```

Worksheet

Query Builder

```
SELECT
    dept.department_id,
    dept.department_name,
    loc.city,
    ctry.country_name,
    reg.region_name
FROM departments dept
JOIN locations loc
    ON dept.location_id = loc.location_id
JOIN countries ctry
    ON loc.country_id = ctry.country_id
JOIN regions reg
    ON ctry.region_id = reg.region_id
ORDER BY dept.department_id;
```

Script Output x

Query Result x

SQL | All Rows Fetched: 27 in 0.015 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME	CITY	COUNTRY_NAME	REGION_NAME
1	10	Administration	Seattle	United States of America	Americas
2	20	Marketing	Toronto	Canada	Americas
3	30	Purchasing	Seattle	United States of America	Americas
4	40	Human Resources	London	United Kingdom	Europe
5	50	Shipping	South San Francisco	United States of America	Americas
6	60	IT	Southlake	United States of America	Americas
7	70	Public Relations	Munich	Germany	Europe
8	80	Sales	Oxford	United Kingdom	Europe
9	90	Executive	Seattle	United States of America	Americas
10	100	Finance	Seattle	United States of America	Americas
11	110	Accounting	Seattle	United States of America	Americas
12	120	Treasury	Seattle	United States of America	Americas
13	130	Corporate Tax	Seattle	United States of America	Americas
14	140	Control And Credit	Seattle	United States of America	Americas
15	150	Shareholder Services	Seattle	United States of America	Americas
16	160	Benefits	Seattle	United States of America	Americas
17	170	Manufacturing	Seattle	United States of America	Americas
18	180	Construction	Seattle	United States of America	Americas
19	190	Contracting	Seattle	United States of America	Americas
20	200	Operations	Seattle	United States of America	Americas

Explanation

This report helps management understand where each department is located.

Query 7: Employee Salary and Job Report

Business requirement

Show employees with salary and job title.

Correct SQL

```
SELECT
```

```
    emp.employee_id,
```

```
    emp.first_name,
```

```
    emp.last_name,
```

```
    job.job_title,
```

```
    emp.salary
```

```
FROM employees emp
```

```
JOIN jobs job
```

```
    ON emp.job_id = job.job_id
```

```
ORDER BY emp.salary DESC;
```

Worksheet

Query Builder

```
SELECT
    emp.employee_id,
    emp.first_name,
    emp.last_name,
    job.job_title,
    emp.salary
FROM employees emp
JOIN jobs job
    ON emp.job_id = job.job_id
ORDER BY emp.salary DESC;
```

Script Output x

Query Result x

SQL | Fetched 50 rows in 0.014 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	JOB_TITLE	SALARY
1	100	Steven	King	President	24000
2	102	Lex	De Haan	Administration Vice President	17000
3	101	Neena	Kochhar	Administration Vice President	17000
4	145	John	Russell	Sales Manager	14000
5	146	Karen	Partners	Sales Manager	13500
6	201	Michael	Hartstein	Marketing Manager	13000
7	108	Nancy	Greenberg	Finance Manager	12008
8	205	Shelley	Higgins	Accounting Manager	12008
9	147	Alberto	Errazuriz	Sales Manager	12000
10	168	Lisa	Ozer	Sales Representative	11500
11	114	Den	Raphaely	Purchasing Manager	11000
12	174	Ellen	Abel	Sales Representative	11000
13	148	Gerald	Cambrault	Sales Manager	11000
14	149	Eleni	Zlotkey	Sales Manager	10500
15	162	Clara	Vishney	Sales Representative	10500
16	169	Harrison	Bloom	Sales Representative	10000
17	204	Hermann	Baer	Public Relations Representative	10000
18	156	Janette	King	Sales Representative	10000
19	150	Peter	Tucker	Sales Representative	10000
20	170	Taylor	Fox	Sales Representative	9600
21	163	Danielle	Greene	Sales Representative	9500
22	157	Patrick	Sully	Sales Representative	9500

Explanation

This report is useful for HR salary and role analysis.

Query 8: Employees by Region

Business requirement

Show employees with their region name.

Correct SQL

```
SELECT
    emp.employee_id,
    emp.first_name,
    emp.last_name,
    dept.department_name,
    reg.region_name
FROM employees emp
LEFT JOIN departments dept
    ON emp.department_id = dept.department_id
LEFT JOIN locations loc
    ON dept.location_id = loc.location_id
LEFT JOIN countries ctry
    ON loc.country_id = ctry.country_id
LEFT JOIN regions reg
    ON ctry.region_id = reg.region_id
ORDER BY reg.region_name, emp.employee_id;
```

Worksheet

Query Builder

```

SELECT
    emp.employee_id,
    emp.first_name,
    emp.last_name,
    dept.department_name,
    reg.region_name
FROM employees emp
LEFT JOIN departments dept
    ON emp.department_id = dept.department_id
LEFT JOIN locations loc
    ON dept.location_id = loc.location_id
LEFT JOIN countries ctry
    ON loc.country_id = ctry.country_id
LEFT JOIN regions reg
    ON ctry.region_id = reg.region_id
ORDER BY reg.region_name, emp.employee_id;

```

Script Output x

Query Result x

SQL | Fetched 50 rows in 0.009 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME	REGION_NAME
1	100	Steven	King	Executive	Americas
2	101	Neena	Kochhar	Executive	Americas
3	102	Lex	De Haan	Executive	Americas
4	103	Alexander	Hunold	IT	Americas
5	104	Bruce	Ernst	IT	Americas
6	105	David	Austin	IT	Americas
7	106	Valli	Pataballa	IT	Americas
8	107	Diana	Lorentz	IT	Americas
9	108	Nancy	Greenberg	Finance	Americas
10	109	Daniel	Faviet	Finance	Americas
11	110	John	Chen	Finance	Americas
12	111	Ismael	Sciarra	Finance	Americas
13	112	Jose Manuel	Urman	Finance	Americas
14	113	Luis	Popp	Finance	Americas
15	114	Den	Raphaely	Purchasing	Americas
16	115	Alexander	Khoo	Purchasing	Americas
17	116	Shelli	Baida	Purchasing	Americas
18	117	Sigal	Tobias	Purchasing	Americas

Explanation

This helps the company report employee distribution by geographic region.

2. Employee & Department Reports

1. Employee ID, Full Name, and Department Name

Employees with departments only

```
SELECT
    emp.employee_id,
    emp.first_name || ' ' || emp.last_name AS full_name,
    dept.department_name
FROM employees emp
JOIN departments dept
    ON emp.department_id = dept.department_id
ORDER BY emp.employee_id;
```

Worksheet

Query Builder

```
SELECT
    emp.employee_id,
    emp.first_name || ' ' || emp.last_name AS full_name,
    dept.department_name
FROM employees emp
JOIN departments dept
    ON emp.department_id = dept.department_id
ORDER BY emp.employee_id;
```

Script Output x

Query Result x

SQL | Fetched 50 rows in 0.005 seconds

	EMPLOYEE_ID	FULL_NAME	DEPARTMENT_NAME
1	100	Steven King	Executive
2	101	Neena Kochhar	Executive
3	102	Lex De Haan	Executive
4	103	Alexander Hunold	IT
5	104	Bruce Ernst	IT
6	105	David Austin	IT
7	106	Valli Pataballa	IT
8	107	Diana Lorentz	IT
9	108	Nancy Greenberg	Finance
10	109	Daniel Faviet	Finance
11	110	John Chen	Finance
12	111	Ismael Sciarra	Finance
13	112	Jose Manuel Urman	Finance
14	113	Luis Popp	Finance
15	114	Den Raphaely	Purchasing
16	115	Alexander Khoo	Purchasing
17	116	Shelli Baida	Purchasing
18	117	Sigal Tobias	Purchasing
19	118	Guy Himuro	Purchasing
20	119	Karen Colmenares	Purchasing
21	120	Matthew Weiss	Shipping
22	121	Adam Fripp	Shipping
23	122	Payam Kaufling	Shipping
24	123	Shanta Vollman	Shipping

2. Employee Full Name, Job Title, and Salary

Employees with valid jobs only

SELECT

```
emp.first_name || ' ' || emp.last_name AS full_name,  
job.job_title,  
emp.salary  
FROM employees emp  
JOIN jobs job  
ON emp.job_id = job.job_id  
ORDER BY emp.salary DESC;
```

Worksheet		Query Builder	
		<pre> SELECT emp.first_name ' ' emp.last_name AS full_name, job.job_title, emp.salary FROM employees emp JOIN jobs job ON emp.job_id = job.job_id ORDER BY emp.salary DESC; </pre>	
Script Output x		Query Result x	
		 Fetched 50 rows in 0.007 seconds	
	FULL_NAME	JOB_TITLE	SALARY
1	Steven King	President	24000
2	Lex De Haan	Administration Vice President	17000
3	Neena Kochhar	Administration Vice President	17000
4	John Russell	Sales Manager	14000
5	Karen Partners	Sales Manager	13500
6	Michael Hartstein	Marketing Manager	13000
7	Nancy Greenberg	Finance Manager	12008
8	Shelley Higgins	Accounting Manager	12008
9	Alberto Errazuriz	Sales Manager	12000
10	Lisa Ozer	Sales Representative	11500
11	Den Raphaely	Purchasing Manager	11000
12	Ellen Abel	Sales Representative	11000
13	Gerald Cambrault	Sales Manager	11000
14	Eleni Zlotkey	Sales Manager	10500
15	Clara Vishney	Sales Representative	10500
16	Harrison Bloom	Sales Representative	10000
17	Hermann Baer	Public Relations Representative	10000
18	Janette King	Sales Representative	10000
19	Peter Tucker	Sales Representative	10000
20	Tayler Fox	Sales Representative	9600
21	Danielle Greene	Sales Representative	9500
22	Patrick Sully	Sales Representative	9500
23	David Bernstein	Sales Representative	9500
24	Allan McEwen	Sales Representative	9000

3. Employee, Department, and Department City

SELECT

emp.first_name || ' ' || emp.last_name AS full_name,

```
dept.department_name,  
loc.city  
FROM employees emp  
JOIN departments dept  
ON emp.department_id = dept.department_id  
JOIN locations loc  
ON dept.location_id = loc.location_id  
ORDER BY emp.employee_id;
```

Worksheet

Query Builder

```

SELECT
    emp.first_name || ' ' || emp.last_name AS full_name,
    dept.department_name,
    loc.city
FROM employees emp
JOIN departments dept
    ON emp.department_id = dept.department_id
JOIN locations loc
    ON dept.location_id = loc.location_id
ORDER BY emp.employee_id;

```

Script Output x

Query Result x

SQL | Fetched 50 rows in 0.008 seconds

	FULL_NAME	DEPARTMENT_NAME	CITY
1	Steven King	Executive	Seattle
2	Neena Kochhar	Executive	Seattle
3	Lex De Haan	Executive	Seattle
4	Alexander Hunold	IT	Southlake
5	Bruce Ernst	IT	Southlake
6	David Austin	IT	Southlake
7	Valli Pataballa	IT	Southlake
8	Diana Lorentz	IT	Southlake
9	Nancy Greenberg	Finance	Seattle
10	Daniel Faviet	Finance	Seattle
11	John Chen	Finance	Seattle
12	Ismael Sciarra	Finance	Seattle
13	Jose Manuel Urman	Finance	Seattle
14	Luis Popp	Finance	Seattle
15	Den Raphaely	Purchasing	Seattle
16	Alexander Khoo	Purchasing	Seattle
17	Shelli Baida	Purchasing	Seattle
18	Sigal Tobias	Purchasing	Seattle
19	Guy Himuro	Purchasing	Seattle
20	Karen Colmenares	Purchasing	Seattle
21	Matthew Weiss	Shipping	South San Francisco
22	Adam Fripp	Shipping	South San Francisco

4. Employee, Department, Country, and Region

SELECT

```
emp.first_name || ' ' || emp.last_name AS full_name,
```

```
dept.department_name,  
ctry.country_name,  
reg.region_name  
FROM employees emp  
JOIN departments dept  
    ON emp.department_id = dept.department_id  
JOIN locations loc  
    ON dept.location_id = loc.location_id  
JOIN countries ctry  
    ON loc.country_id = ctry.country_id  
JOIN regions reg  
    ON ctry.region_id = reg.region_id  
ORDER BY emp.employee_id;
```

Worksheet

Query Builder

```

SELECT
  emp.first_name || ' ' || emp.last_name AS full_name,
  dept.department_name,
  ctry.country_name,
  reg.region_name
FROM employees emp
JOIN departments dept
  ON emp.department_id = dept.department_id
JOIN locations loc
  ON dept.location_id = loc.location_id
JOIN countries ctry
  ON loc.country_id = ctry.country_id
JOIN regions reg
  ON ctry.region_id = reg.region_id
ORDER BY emp.employee_id;

```

Script Output x

Query Result x

SQL | Fetched 50 rows in 0.01 seconds

	FULL_NAME	DEPARTMENT_NAME	COUNTRY_NAME	REGION_NAME
1	Steven King	Executive	United States of America	Americas
2	Neena Kochhar	Executive	United States of America	Americas
3	Lex De Haan	Executive	United States of America	Americas
4	Alexander Hunold	IT	United States of America	Americas
5	Bruce Ernst	IT	United States of America	Americas
6	David Austin	IT	United States of America	Americas
7	Valli Pataballa	IT	United States of America	Americas
8	Diana Lorentz	IT	United States of America	Americas
9	Nancy Greenberg	Finance	United States of America	Americas
10	Daniel Faviet	Finance	United States of America	Americas
11	John Chen	Finance	United States of America	Americas
12	Ismael Sciarra	Finance	United States of America	Americas
13	Jose Manuel Urman	Finance	United States of America	Americas
14	Luis Popp	Finance	United States of America	Americas
15	Den Raphaely	Purchasing	United States of America	Americas
16	Alexander Khoo	Purchasing	United States of America	Americas
17	Shelli Baida	Purchasing	United States of America	Americas
18	Sigal Tobias	Purchasing	United States of America	Americas

3. Organization Structure Analysis

5. Complete employee list including employees without departments

SELECT

emp.employee_id,

```
emp.first_name || ' ' || emp.last_name AS full_name,  
dept.department_name  
FROM employees emp  
LEFT JOIN departments dept  
ON emp.department_id = dept.department_id  
ORDER BY emp.employee_id;
```

Worksheet		Query Builder	
		<pre> SELECT emp.employee_id, emp.first_name ' ' emp.last_name AS full_name, dept.department_name FROM employees emp LEFT JOIN departments dept ON emp.department_id = dept.department_id ORDER BY emp.employee_id; </pre>	
Script Output x		Query Result x	
		 Fetched 50 rows in 0.007 seconds	
	EMPLOYEE_ID	FULL_NAME	DEPARTMENT_NAME
1	100	Steven King	Executive
2	101	Neena Kochhar	Executive
3	102	Lex De Haan	Executive
4	103	Alexander Hunold	IT
5	104	Bruce Ernst	IT
6	105	David Austin	IT
7	106	Valli Pataballa	IT
8	107	Diana Lorentz	IT
9	108	Nancy Greenberg	Finance
10	109	Daniel Faviet	Finance
11	110	John Chen	Finance
12	111	Ismael Sciarra	Finance
13	112	Jose Manuel Urman	Finance
14	113	Luis Popp	Finance
15	114	Den Raphaely	Purchasing
16	115	Alexander Khoo	Purchasing
17	116	Shelli Baida	Purchasing
18	117	Sigal Tobias	Purchasing
19	118	Guy Himuro	Purchasing
20	119	Karen Colmenares	Purchasing
21	120	Matthew Weiss	Shipping
22	121	Adam Fripp	Shipping
23	122	Payam Kaufling	Shipping
24	123	Shanta Vollman	Shipping
25	124	Kevin Mourgous	Shipping

Reason:

LEFT JOIN keeps all employees, even if department_id is NULL.

6. Complete department list including departments without employees

```
SELECT
    dept.department_id,
    dept.department_name,
    emp.employee_id,
    emp.first_name || ' ' || emp.last_name AS employee_name
FROM departments dept
LEFT JOIN employees emp
    ON dept.department_id = emp.department_id
ORDER BY dept.department_id, emp.employee_id;
```

Worksheet

Query Builder

```
SELECT
dept.department_id,
dept.department_name,
emp.employee_id,
emp.first_name || ' ' || emp.last_name AS employee_name
FROM departments dept
LEFT JOIN employees emp
ON dept.department_id = emp.department_id
ORDER BY dept.department_id, emp.employee_id;
```

Script Output x

Query Result x

SQL | Fetched 50 rows in 0.007 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME	EMPLOYEE_ID	EMPLOYEE_NAME
1	10	Administration	200	Jennifer Whalen
2	20	Marketing	201	Michael Hartstein
3	20	Marketing	202	Pat Fay
4	30	Purchasing	114	Den Raphaely
5	30	Purchasing	115	Alexander Khoo
6	30	Purchasing	116	Shelli Baida
7	30	Purchasing	117	Sigal Tobias
8	30	Purchasing	118	Guy Himuro
9	30	Purchasing	119	Karen Colmenares
10	40	Human Resources	203	Susan Mavris
11	50	Shipping	120	Matthew Weiss
12	50	Shipping	121	Adam Fripp
13	50	Shipping	122	Payam Kaufling
14	50	Shipping	123	Shanta Vollman
15	50	Shipping	124	Kevin Mourgous
16	50	Shipping	125	Julia Nayer
17	50	Shipping	126	Irene Mikkilineni
18	50	Shipping	127	James Landry
19	50	Shipping	128	Steven Markle
20	50	Shipping	129	Laura Bissot
21	50	Shipping	130	Mozhe Atkinson
22	50	Shipping	131	James Marlow
23	50	Shipping	132	TJ Olson
24	50	Shipping	133	Jason Mallin
25	50	Shipping	134	Michael Rogers

Reason:

LEFT JOIN starts from departments, so all departments appear even if they have no employees.

7. Departments that currently have no employees assigned

```
SELECT
    dept.department_id,
    dept.department_name
FROM departments dept
LEFT JOIN employees emp
    ON dept.department_id = emp.department_id
WHERE emp.employee_id IS NULL
ORDER BY dept.department_id;
```

Worksheet		Query Builder																																																			
		<pre> SELECT dept.department_id, dept.department_name FROM departments dept LEFT JOIN employees emp ON dept.department_id = emp.department_id WHERE emp.employee_id IS NULL ORDER BY dept.department_id; </pre>																																																			
		Script Output x Query Result x SQL All Rows Fetched: 16 in 0.005 seconds <table> <thead> <tr> <th></th><th>DEPARTMENT_ID</th><th>DEPARTMENT_NAME</th></tr> </thead> <tbody> <tr><td>1</td><td>120</td><td>Treasury</td></tr> <tr><td>2</td><td>130</td><td>Corporate Tax</td></tr> <tr><td>3</td><td>140</td><td>Control And Credit</td></tr> <tr><td>4</td><td>150</td><td>Shareholder Services</td></tr> <tr><td>5</td><td>160</td><td>Benefits</td></tr> <tr><td>6</td><td>170</td><td>Manufacturing</td></tr> <tr><td>7</td><td>180</td><td>Construction</td></tr> <tr><td>8</td><td>190</td><td>Contracting</td></tr> <tr><td>9</td><td>200</td><td>Operations</td></tr> <tr><td>10</td><td>210</td><td>IT Support</td></tr> <tr><td>11</td><td>220</td><td>NOC</td></tr> <tr><td>12</td><td>230</td><td>IT Helpdesk</td></tr> <tr><td>13</td><td>240</td><td>Government Sales</td></tr> <tr><td>14</td><td>250</td><td>Retail Sales</td></tr> <tr><td>15</td><td>260</td><td>Recruiting</td></tr> <tr><td>16</td><td>270</td><td>Payroll</td></tr> </tbody> </table>		DEPARTMENT_ID	DEPARTMENT_NAME	1	120	Treasury	2	130	Corporate Tax	3	140	Control And Credit	4	150	Shareholder Services	5	160	Benefits	6	170	Manufacturing	7	180	Construction	8	190	Contracting	9	200	Operations	10	210	IT Support	11	220	NOC	12	230	IT Helpdesk	13	240	Government Sales	14	250	Retail Sales	15	260	Recruiting	16	270	Payroll
	DEPARTMENT_ID	DEPARTMENT_NAME																																																			
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10	210	IT Support																																																			
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12	230	IT Helpdesk																																																			
13	240	Government Sales																																																			
14	250	Retail Sales																																																			
15	260	Recruiting																																																			
16	270	Payroll																																																			

Reason:

After the **LEFT JOIN**, departments with no matching employees have **NULL** employee values.

4. Management Hierarchy Analysis

8. Employee name and direct manager name

SELECT

emp.first_name || ' ' || emp.last_name AS employee_name,

```
mgr.first_name || ' ' || mgr.last_name AS manager_name
FROM employees emp
LEFT JOIN employees mgr
  ON emp.manager_id = mgr.employee_id
ORDER BY emp.employee_id;
```

Worksheet

Query Builder

```

SELECT
  emp.first_name || ' ' || emp.last_name AS employee_name,
  mgr.first_name || ' ' || mgr.last_name AS manager_name
FROM employees emp
LEFT JOIN employees mgr
  ON emp.manager_id = mgr.employee_id
ORDER BY emp.employee_id;

```

Script Output x

Query Result x





 | Fetched 50 rows in 0.008 seconds

	EMPLOYEE_NAME	MANAGER_NAME
1	Steven King	
2	Neena Kochhar	Steven King
3	Lex De Haan	Steven King
4	Alexander Hunold	Lex De Haan
5	Bruce Ernst	Alexander Hunold
6	David Austin	Alexander Hunold
7	Valli Pataballa	Alexander Hunold
8	Diana Lorentz	Alexander Hunold
9	Nancy Greenberg	Neena Kochhar
10	Daniel Faviet	Nancy Greenberg
11	John Chen	Nancy Greenberg
12	Ismael Sciarra	Nancy Greenberg
13	Jose Manuel Urman	Nancy Greenberg
14	Luis Popp	Nancy Greenberg
15	Den Raphaely	Steven King
16	Alexander Khoo	Den Raphaely
17	Shelli Baida	Den Raphaely
18	Sigal Tobias	Den Raphaely
19	Guy Himuro	Den Raphaely
20	Karen Colmenares	Den Raphaely
21	Matthew Weiss	Steven King
22	Adam Fripp	Steven King
23	Payam Kaufling	Steven King
24	Shanta Vollman	Steven King

Reason:

This is a **self join** because the manager is also stored in the **employees** table.

Correct relationship:

emp.manager_id = mgr.employee_id

9. Employees without managers

SELECT

emp.employee_id,

emp.first_name || ' ' || emp.last_name AS employee_name

FROM employees emp

WHERE emp.manager_id IS NULL

ORDER BY emp.employee_id;

The screenshot shows a SQL query builder interface with a 'Query Builder' tab. The query is as follows:

```
SELECT
    emp.employee_id,
    emp.first_name || ' ' || emp.last_name AS employee_name
FROM employees emp
WHERE emp.manager_id IS NULL
ORDER BY emp.employee_id;
```

Below the query editor, the 'Query Result' tab is active, showing the results of the query. The status bar indicates 'All Rows Fetched: 1 in 0.004 seconds'. The results are displayed in a table with two columns: 'EMPLOYEE_ID' and 'EMPLOYEE_NAME'.

EMPLOYEE_ID	EMPLOYEE_NAME
1	100 Steven King

Brief explanation:

Employees without managers usually represent the **top-level manager** in the organization, such as the company president or CEO. Their **manager_id** is **NULL** because no one is above them in the hierarchy.

5. Advanced Business Analysis

10. Employees earning more than their managers

SELECT

emp.employee_id,

emp.first_name || ' ' || emp.last_name AS employee_name,

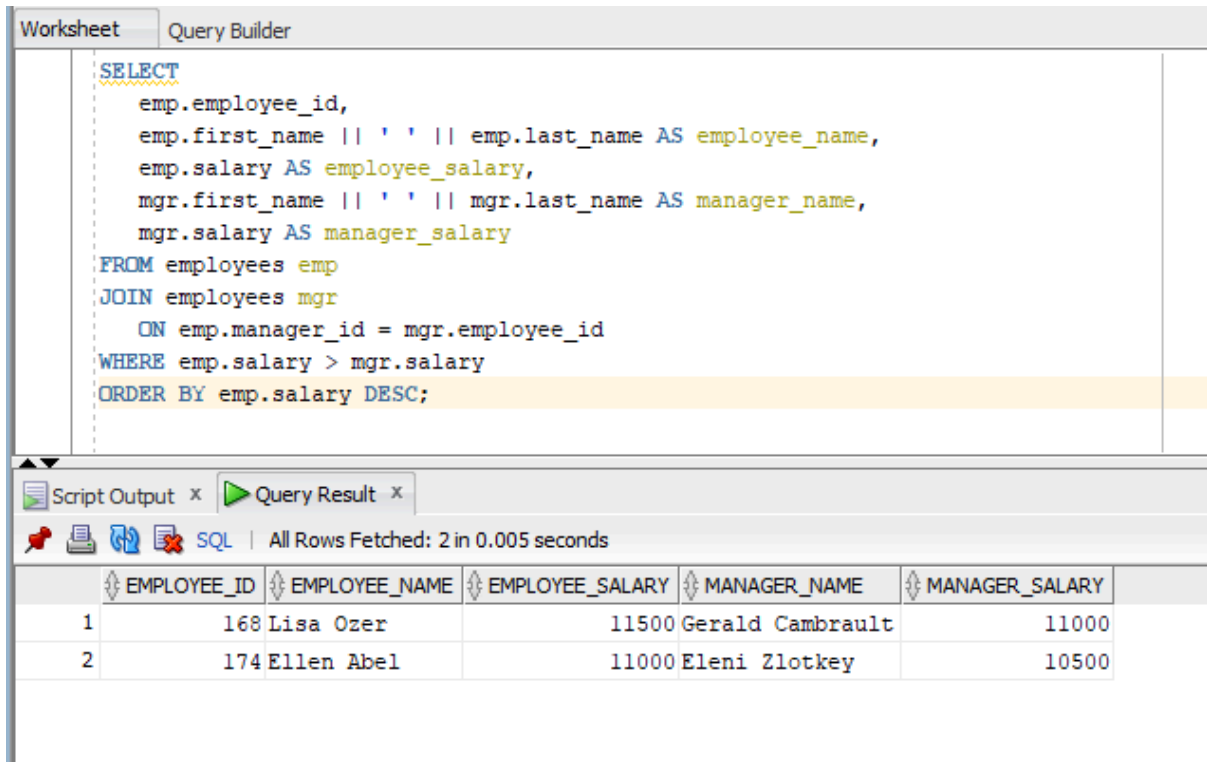
emp.salary AS employee_salary,

mgr.first_name || ' ' || mgr.last_name AS manager_name,

```

mgr.salary AS manager_salary
FROM employees emp
JOIN employees mgr
  ON emp.manager_id = mgr.employee_id
WHERE emp.salary > mgr.salary
ORDER BY emp.salary DESC;

```



The screenshot shows a SQL query in a 'Query Builder' window. The query is as follows:

```

SELECT
  emp.employee_id,
  emp.first_name || ' ' || emp.last_name AS employee_name,
  emp.salary AS employee_salary,
  mgr.first_name || ' ' || mgr.last_name AS manager_name,
  mgr.salary AS manager_salary
FROM employees emp
JOIN employees mgr
  ON emp.manager_id = mgr.employee_id
WHERE emp.salary > mgr.salary
ORDER BY emp.salary DESC;

```

Below the query editor, the 'Query Result' window displays the results of the query. It shows two rows of data:

	EMPLOYEE_ID	EMPLOYEE_NAME	EMPLOYEE_SALARY	MANAGER_NAME	MANAGER_SALARY
1	168	Lisa Ozer	11500	Gerald Cambrault	11000
2	174	Ellen Abel	11000	Eleni Zlotkey	10500

Reason:

This uses a self join. It compares each employee's salary with their direct manager's salary.

11. Departments where the department manager works within the same department

```

SELECT
  dept.department_id,
  dept.department_name,
  mgr.first_name || ' ' || mgr.last_name AS department_manager
FROM departments dept
JOIN employees mgr
  ON dept.manager_id = mgr.employee_id
WHERE dept.department_id = mgr.department_id
ORDER BY dept.department_id;

```

Worksheet

Query Builder

```

SELECT
    dept.department_id,
    dept.department_name,
    mgr.first_name || ' ' || mgr.last_name AS department_manager
FROM departments dept
JOIN employees mgr
    ON dept.manager_id = mgr.employee_id
WHERE dept.department_id = mgr.department_id
ORDER BY dept.department_id;

```

Script Output x

Query Result x

SQL | All Rows Fetched: 11 in 0.059 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME	DEPARTMENT_MANAGER
1	10	Administration	Jennifer Whalen
2	20	Marketing	Michael Hartstein
3	30	Purchasing	Den Raphaely
4	40	Human Resources	Susan Mavris
5	50	Shipping	Adam Fripp
6	60	IT	Alexander Hunold
7	70	Public Relations	Hermann Baer
8	80	Sales	John Russell
9	90	Executive	Steven King
10	100	Finance	Nancy Greenberg
11	110	Accounting	Shelley Higgins

Reason:

The department manager is stored in `departments.manager_id`.

This query checks that the manager's own `department_id` is the same department they manage.

6. Executive Dashboard Reports

12. Employee, job title, department, city, and country

```

SELECT
    emp.employee_id,
    emp.first_name || ' ' || emp.last_name AS employee_name,
    job.job_title,
    dept.department_name,
    loc.city,
    ctry.country_name
FROM employees emp

```

```

JOIN jobs job
  ON emp.job_id = job.job_id
LEFT JOIN departments dept
  ON emp.department_id = dept.department_id
LEFT JOIN locations loc
  ON dept.location_id = loc.location_id
LEFT JOIN countries ctry
  ON loc.country_id = ctry.country_id
ORDER BY emp.employee_id;

```

Worksheet

Query Builder

```
SELECT
    emp.employee_id,
    emp.first_name || ' ' || emp.last_name AS employee_name,
    job.job_title,
    dept.department_name,
    loc.city,
    ctry.country_name
FROM employees emp
JOIN jobs job
    ON emp.job_id = job.job_id
LEFT JOIN departments dept
    ON emp.department_id = dept.department_id
LEFT JOIN locations loc
    ON dept.location_id = loc.location_id
LEFT JOIN countries ctry
    ON loc.country_id = ctry.country_id
ORDER BY emp.employee_id;
```

Script Output x Query Result x

SQL | Fetched 50 rows in 0.044 seconds

	EMPLOYEE_ID	EMPLOYEE_NAME	JOB_TITLE	DEPARTMENT_NAME	CITY	COUNTRY_NAME
1	100	Steven King	President	Executive	Seattle	United States of America
2	101	Neena Kochhar	Administration Vice President	Executive	Seattle	United States of America
3	102	Lex De Haan	Administration Vice President	Executive	Seattle	United States of America
4	103	Alexander Hunold	Programmer	IT	Southlake	United States of America
5	104	Bruce Ernst	Programmer	IT	Southlake	United States of America
6	105	David Austin	Programmer	IT	Southlake	United States of America
7	106	Valli Pataballa	Programmer	IT	Southlake	United States of America
8	107	Diana Lorentz	Programmer	IT	Southlake	United States of America
9	108	Nancy Greenberg	Finance Manager	Finance	Seattle	United States of America
10	109	Daniel Faviet	Accountant	Finance	Seattle	United States of America
11	110	John Chen	Accountant	Finance	Seattle	United States of America
12	111	Ismael Sciarra	Accountant	Finance	Seattle	United States of America
13	112	Jose Manuel Urman	Accountant	Finance	Seattle	United States of America
14	113	Luis Popp	Accountant	Finance	Seattle	United States of America
15	114	Den Raphaely	Purchasing Manager	Purchasing	Seattle	United States of America
16	115	Alexander Khoo	Purchasing Clerk	Purchasing	Seattle	United States of America
17	116	Shelli Baida	Purchasing Clerk	Purchasing	Seattle	United States of America

13. Department, manager, city, and country

Including departments without managers

```

SELECT
  dept.department_id,
  dept.department_name,
  mgr.first_name || ' ' || mgr.last_name AS manager_name,
  loc.city,
  ctry.country_name
FROM departments dept

```

LEFT JOIN employees mgr
 ON dept.manager_id = mgr.employee_id
 LEFT JOIN locations loc
 ON dept.location_id = loc.location_id
 LEFT JOIN countries ctry
 ON loc.country_id = ctry.country_id
 ORDER BY dept.department_id;

Worksheet

Query Builder

```
SELECT
dept.department_id,
dept.department_name,
mgr.first_name || ' ' || mgr.last_name AS manager_name,
loc.city,
ctry.country_name
FROM departments dept
LEFT JOIN employees mgr
  ON dept.manager_id = mgr.employee_id
LEFT JOIN locations loc
  ON dept.location_id = loc.location_id
LEFT JOIN countries ctry
  ON loc.country_id = ctry.country_id
ORDER BY dept.department_id;
```

Script Output x

Query Result x

SQL | All Rows Fetched: 27 in 0.014 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_NAME	CITY	COUNTRY_NAME
1	10	Administration	Jennifer Whalen	Seattle	United States of America
2	20	Marketing	Michael Hartstein	Toronto	Canada
3	30	Purchasing	Den Raphaely	Seattle	United States of America
4	40	Human Resources	Susan Mavris	London	United Kingdom
5	50	Shipping	Adam Fripp	South San Francisco	United States of America
6	60	IT	Alexander Hunold	Southlake	United States of America
7	70	Public Relations	Hermann Baer	Munich	Germany
8	80	Sales	John Russell	Oxford	United Kingdom
9	90	Executive	Steven King	Seattle	United States of America
10	100	Finance	Nancy Greenberg	Seattle	United States of America
11	110	Accounting	Shelley Higgins	Seattle	United States of America
12	120	Treasury		Seattle	United States of America
13	130	Corporate Tax		Seattle	United States of America
14	140	Control And Credit		Seattle	United States of America
15	150	Shareholder Services		Seattle	United States of America
16	160	Benefits		Seattle	United States of America
17	170	Manufacturing		Seattle	United States of America

14. Employees working in the same department as their manager

SELECT
 emp.employee_id,
 emp.first_name || ' ' || emp.last_name AS employee_name,
 dept.department_name,
 mgr.first_name || ' ' || mgr.last_name AS manager_name
 FROM employees emp

```

JOIN employees mgr
  ON emp.manager_id = mgr.employee_id
JOIN departments dept
  ON emp.department_id = dept.department_id
WHERE emp.department_id = mgr.department_id
ORDER BY emp.employee_id;

```

Worksheet

Query Builder

```

SELECT
  emp.employee_id,
  emp.first_name || ' ' || emp.last_name AS employee_name,
  dept.department_name,
  mgr.first_name || ' ' || mgr.last_name AS manager_name
FROM employees emp
JOIN employees mgr
  ON emp.manager_id = mgr.employee_id
JOIN departments dept
  ON emp.department_id = dept.department_id
WHERE emp.department_id = mgr.department_id
ORDER BY emp.employee_id;

```

Script Output x

Query Result x

SQL | Fetched 50 rows in 0.009 seconds

	EMPLOYEE_ID	EMPLOYEE_NAME	DEPARTMENT_NAME	MANAGER_NAME
1	101	Neena Kochhar	Executive	Steven King
2	102	Lex De Haan	Executive	Steven King
3	104	Bruce Ernst	IT	Alexander Hunold
4	105	David Austin	IT	Alexander Hunold
5	106	Valli Pataballa	IT	Alexander Hunold
6	107	Diana Lorentz	IT	Alexander Hunold
7	109	Daniel Faviet	Finance	Nancy Greenberg
8	110	John Chen	Finance	Nancy Greenberg
9	111	Ismael Sciarra	Finance	Nancy Greenberg
10	112	Jose Manuel Urman	Finance	Nancy Greenberg
11	113	Luis Popp	Finance	Nancy Greenberg
12	115	Alexander Khoo	Purchasing	Den Raphaely
13	116	Shelli Baida	Purchasing	Den Raphaely
14	117	Sigal Tobias	Purchasing	Den Raphaely
15	118	Guy Himuro	Purchasing	Den Raphaely
16	119	Karen Colmenares	Purchasing	Den Raphaely
17	125	Julia Nayer	Shipping	Matthew Weiss

15. Employees whose managers work in different departments

```
SELECT
    emp.employee_id,
    emp.first_name || ' ' || emp.last_name AS employee_name,
    emp_dept.department_name AS employee_department,
    mgr.first_name || ' ' || mgr.last_name AS manager_name,
    mgr_dept.department_name AS manager_department
FROM employees emp
JOIN employees mgr
    ON emp.manager_id = mgr.employee_id
LEFT JOIN departments emp_dept
    ON emp.department_id = emp_dept.department_id
LEFT JOIN departments mgr_dept
    ON mgr.department_id = mgr_dept.department_id
WHERE emp.department_id <> mgr.department_id
ORDER BY emp.employee_id;
```

Worksheet

Query Builder

```

SELECT
  emp.employee_id,
  emp.first_name || ' ' || emp.last_name AS employee_name,
  emp_dept.department_name AS employee_department,
  mgr.first_name || ' ' || mgr.last_name AS manager_name,
  mgr_dept.department_name AS manager_department
FROM employees emp
JOIN employees mgr
  ON emp.manager_id = mgr.employee_id
LEFT JOIN departments emp_dept
  ON emp.department_id = emp_dept.department_id
LEFT JOIN departments mgr_dept
  ON mgr.department_id = mgr_dept.department_id
WHERE emp.department_id <> mgr.department_id
ORDER BY emp.employee_id;

```

Script Output x

Query Result x

All Rows Fetched: 18 in 0.042 seconds

	EMPLOYEE_ID	EMPLOYEE_NAME	EMPLOYEE_DEPARTMENT	MANAGER_NAME	MANAGER_DEPARTMENT
1	103	Alexander Hunold	IT	Lex De Haan	Executive
2	108	Nancy Greenberg	Finance	Neena Kochhar	Executive
3	114	Den Raphaely	Purchasing	Steven King	Executive
4	120	Matthew Weiss	Shipping	Steven King	Executive
5	121	Adam Fripp	Shipping	Steven King	Executive
6	122	Payam Kaufling	Shipping	Steven King	Executive
7	123	Shanta Vollman	Shipping	Steven King	Executive
8	124	Kevin Mourgous	Shipping	Steven King	Executive
9	145	John Russell	Sales	Steven King	Executive
10	146	Karen Partners	Sales	Steven King	Executive
11	147	Alberto Errazuriz	Sales	Steven King	Executive
12	148	Gerald Cambrault	Sales	Steven King	Executive
13	149	Eleni Zlotkey	Sales	Steven King	Executive
14	200	Jennifer Whalen	Administration	Neena Kochhar	Executive
15	201	Michael Hartstein	Marketing	Steven King	Executive
16	203	Susan Mavris	Human Resources	Neena Kochhar	Executive
17	204	Hermann Baer	Public Relations	Neena Kochhar	Executive

16. Distinct countries containing company departments

```

SELECT DISTINCT
  ctry.country_name
FROM departments dept
JOIN locations loc
  ON dept.location_id = loc.location_id
JOIN countries ctry
  ON loc.country_id = ctry.country_id
ORDER BY ctry.country_name;

```

The screenshot shows a SQL Query Builder window with two tabs: 'Worksheet' and 'Query Builder'. The 'Query Builder' tab is active, displaying the following SQL query:

```
SELECT DISTINCT
  ctry.country_name
FROM departments dept
JOIN locations loc
  ON dept.location_id = loc.location_id
JOIN countries ctry
  ON loc.country_id = ctry.country_id
ORDER BY ctry.country_name;
```

Below the query editor, there is a 'Script Output' tab and a 'Query Result' tab. The 'Query Result' tab is active, showing the results of the query. The status bar indicates 'All Rows Fetched: 4 in 0.024 seconds'.

	COUNTRY_NAME
1	Canada
2	Germany
3	United Kingdom
4	United States of America

Brief Notes

Use **LEFT JOIN** when the report must include missing data, such as departments without managers.

Use self join on **employees** when comparing employees with their managers.